03-ELEMENTARY CHARGE and MILLIKAN EXPERIMENT

PURPOSE of the EXPERIMENT

This experiment involves observing charged oil droplets accelerated between two capacitor plates.

1. Measurement of oil droplets' rise and fall times with various charges at different voltages.

2. Determination of the charge and radius of oil droplets.

3. Determination of the elementary charge from the velocity of the oil droplets in the direction of gravity and the opposite direction.

4. Determination of the e/m ratio of the electron.

GENERAL INFORMATION

The increasing and decreasing motion of charged oil droplets in the electric field of the capacitor is observed and their velocity is determined:

The velocity of the fall of the oil droplet in the electric field	υ_1
The velocity of the rise of the oil droplet in the electric field	υ_2
The voltage of the capacitor	U
Charge on droplets	$Q = n.e$
The radius of droplets	r
Capacitor inner-electrode distance	$d = 2.5mm \pm 0.01mm$
Density of silicone oil	$\rho_1 = 1.03.10^3 kgm^{-3}$
Viscosity of air	$\eta = 1.82.10^{-5} kg(m.s)^{-1}$
Acceleration due to gravity	$g = 9.81ms^{-2}$
Density of air	$\rho_2 = 1.293kgm^{-3}$

The force F applied to a sphere of radius r and velocity υ in a viscous, non-fluid liquid of viscosity η:

$$F = 6\,\pi\, r\, \eta\, \upsilon \qquad \text{(Stoke Kanunu).} \qquad (1)$$

A droplet of mass m, volume V, and density ρ_1 is located in the earth's gravitational field.

$$F = m.g = \rho_1.V.g \qquad (2)$$

Buoyancy is given as follows:

$$F = \rho_2 .V.g \qquad V = \frac{4}{3}\pi\, r^3 \tag{3}$$

Electric field force,

$$F = Q.E = Q.\frac{U}{d} \tag{4}$$

From the total forces acting on the charged particle, the velocities of the fall and rise of the droplets are obtained.

$$\upsilon_1 = \frac{1}{6\pi r\eta}\left(Q.\frac{U}{d} + \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right) \tag{5}$$

$$\upsilon_2 = \frac{1}{6\pi r\eta}\left(Q.\frac{U}{d} - \frac{4}{3}\pi r^3 g(\rho_1 - \rho_2)\right) \tag{6}$$

Adding and subtracting these equations gives the charge and radius of the droplet.

$$Q = C_1 \frac{\upsilon_1 + \upsilon_2}{U}\sqrt{|\upsilon_1 - \upsilon_2|} \tag{7}$$

$$C_1 = \frac{9}{2}\pi d\sqrt{\frac{\eta^3}{g(\rho_1 - \rho_2)}}$$

$$C_1 = 2.73.10^{-11} kgm(m.s)^{-1/2}$$

$$r = C_2\sqrt{|\upsilon_1 - \upsilon_2|} \tag{8}$$

$$C_2 = \frac{3}{2}\sqrt{\frac{\eta}{g(\rho_1 - \rho_2)}}$$

$$C_2 = 6.37.10^{-5}(m.s)^{1/2}$$

Figure 1 shows that the charge of the droplets has precise values whose coefficient is the elementary charge *e*.

$$Q = n.e$$

The average value is obtained as the elementary charge $e = 1.68.10^{-19}$ As.

$$kg\sqrt{\frac{m}{s}}\frac{\left(\frac{\mathrm{m}}{\mathrm{s}}\right)}{\mathrm{V}}\sqrt{\frac{\mathrm{m}}{\mathrm{s}}}=kg\frac{m}{s}\frac{\left(\frac{\mathrm{m}}{\mathrm{s}}\right)}{\left(\frac{\mathrm{kg\,m^2}}{\mathrm{s^3\,A}}\right)}=\mathrm{kg}\frac{\mathrm{m}^2}{s^2}\frac{\mathrm{s}^3A}{kg\,\mathrm{m}^2}=As=C$$

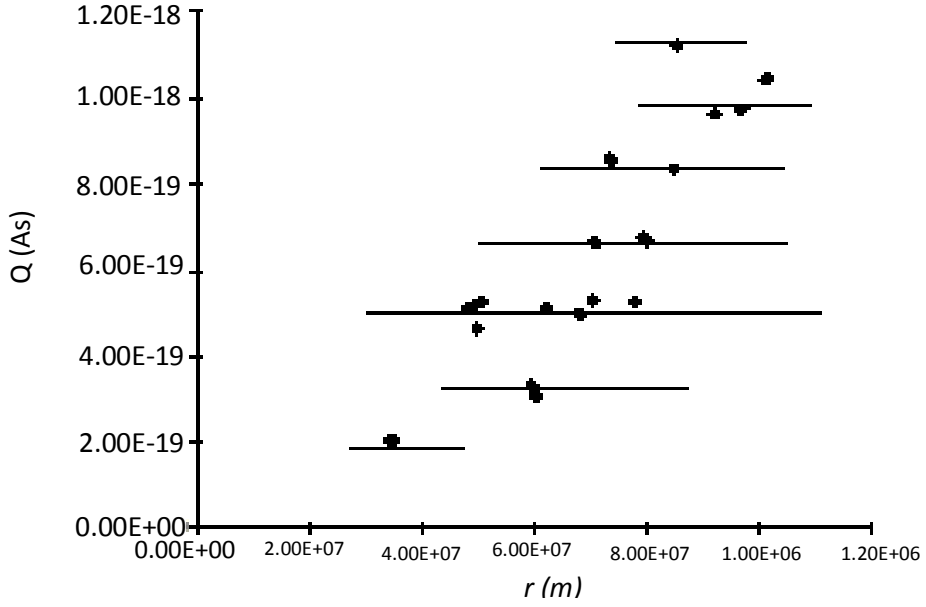


Figure 1: Various droplets measured by the Millikan Method to determine the elementary charge.

HOW TO CONDUCT THE EXPERIMENT

The experimental setup is as follows.

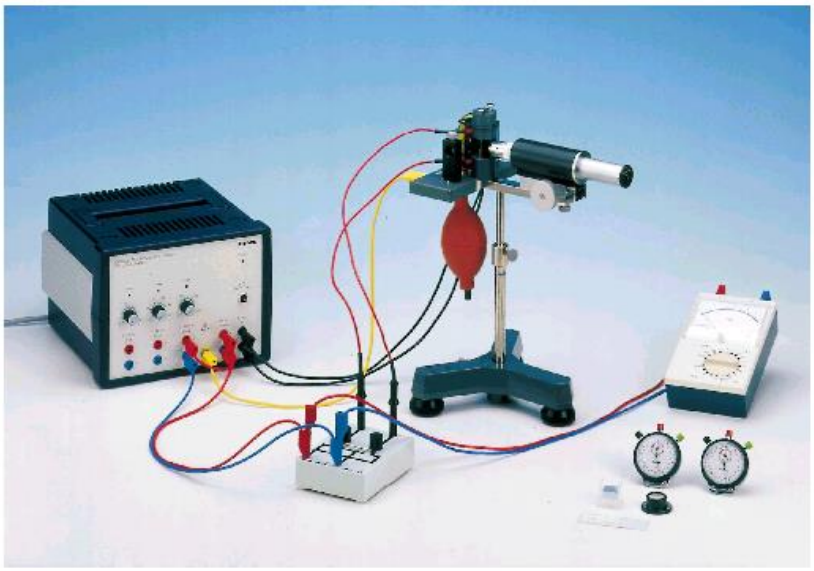


Figure 2: Experimental setup for the determination of the elementary charge with Millikan oil drop method.

Multi-range meter, Power Supply (0...600 V DC), Stage micrometer (1 mm-100 divisions), Stopwatch, Coated Glasses (18318 mm) 50 pcs, Commutator Switch, Stand Tube, Connection Cables.

The microscope's eyepiece micrometer is calibrated with a stage micrometer. The stage micrometer has two small rings in the center and the center of the smallest one is a scale. This scale is used for eyepiece micrometer adjustment.

1. To do this, a stage micrometer is slid into the slits between the microscope lens and the open small chambers (Figure 3).

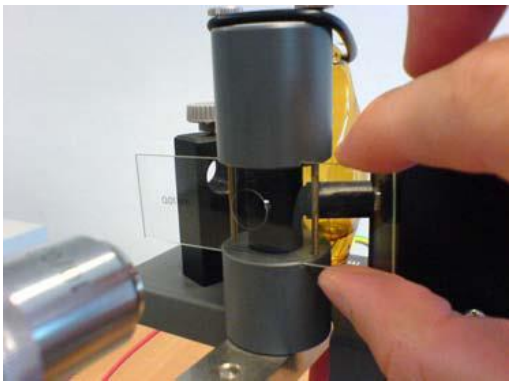


Figure 3

2. Look through the microscope and adjust the focus so that the scale is visible (the two rings around the scale should be used as orientation). The distance between the microscope lens and the stage micrometer should be only a few millimeters.

3. The two scales should overlap.

4. Now count the lines of the stage micrometer, the eyepiece micrometer should be covered by 30 lines.

5. Therefore, convert the 30 divisions of the eyepiece micrometer into mm so that 100 lines on the stage micrometer correspond to 1 mm.

6. Record the calibration result.

7. Remove the stage micrometer from the slits to start the experiment.

8. Sometimes it is useful to put a cover glass in the slits to protect the chamber against drafts.

9. Set the multi-range meter to the 600 V D.C. measurement region.

10. Set the capacitor voltage to a value between 300 V and 500 V. (Set the rotary switch for voltage on the power supply to 0.)

11. Look into the microscope and move some oil droplets with the sprayer several times.

12. Adjust the focus of the microscope until oil droplets like small white rings appear (Figure 4).

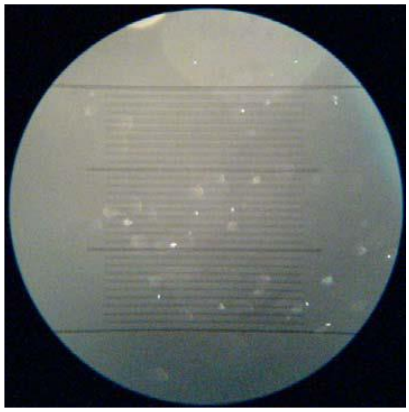


Figure 4

13. Observe the movement of the oil droplet and find its charge. (An oil droplet can be identified as a charge when the polarity of the capacitor is reversed and the orientation is changed by the commander switch setting.)

14. To see the oil droplet for a long time, the microscope must be properly focused.

15. Try moving the charged oil droplet several times between the highest and lowest degrees on the eyepiece micrometer with the setting of the commander switch.

16. When you do not see any oil droplets, spray some droplets again (sometimes it is useful to turn the power supply off and then on again after a few seconds).

The experiment should be carried out by at least two people.

1. Select an oil droplet and use the commutator switch to move the droplet between the highest and lowest degrees on the eyepiece micrometer. Correct the focusing of the microscope if necessary.

Note: Criteria to follow when selecting oil droplets:

- The droplet should not move too fast, it should have a small charge (between 1 and 3 s for 30 divisions).
- The droplet should not move too slowly and should not show excessive movements. If desired, the capacitor voltage is increased.

2. As soon as you find the oil droplet, collect a few ascent times with the first stopwatch (t_1) and a few descent times with the second stopwatch (t_2). Notice that the image you see under the microscope is inverted. Write the results in Table 1.

Note: For both cases the addition times must be greater than 5 seconds.

3. With the help of the calibration, convert the distances traveled by the oil droplets into "s" mm and write your results in Table 1.(columns 4 and 7)

4. Repeat the measurement twice, 400 V and 500 V.

Velocity values υ_1 and υ_2 for each of the three oil droplets

$$v = \frac{s}{t}$$

With the above equation, s_1 and t_1 or s_2 and t_2 can be calculated from the measurement results shown in Table 1. Do not forget to convert the s_1 and s_2 results into m units. Write your results below in Table 2.

5. Now write down your results in Table 2 using the equations below and calculate the charges Q along with the radii r of the oil droplets. Find the mean of the result and calculate its error.

$$Q = C_1 \frac{v_1 + v_2}{U} \sqrt{|v_1 - v_2|}$$

$$C_1 = 2.73x10^{-11} kg(m/s)^{-1/2}$$

$$r = C_2 \sqrt{|v_1 - v_2|}$$

$$C_2 = 6.37x10^{-5}(mxs)^{1/2}$$

6. Combine the possible Q charge and r radius results for the oil droplets to decide the elementary charge *e* from the measurement results. Plot the Q charges corresponding to the radius r of the oil droplets (as in Figure 1). Interpret the graph.

7. Now draw the lines defining each charge level. The graph should look like the one in Figure 1.

8. In the next step, determine the distance between each line. These distances correspond to the elementary charge, *e*. Remember the distance between the x-axis (Q=0) and the first line. Calculate the distances and average them.

Calibration 30 divisions = 0.89 mm (to be calculated by calibration)

Table 1: Results

Voltage U[V]	Ascent times t_1[s]	Distance traveled s_1[div]	Distance traveled s_1[mm]	Descent times t_2[s]	Distance traveled s_2[div]	Distance traveled s_2[mm]
300						
300						
300						
400						
400						
400						
500						
500						
500						

Table 2: Results

Voltage U[V]	Ascent velocity v_1 [m/s]	Descent velocity v_2 [m/s]	Charge Q [C]	Radius *r* [m]
300				
300				
300				
400				
400				
400				
500				
500				
500				
Average Q:				

9. Plot the charges Q corresponding to the radius r of the oil droplets.

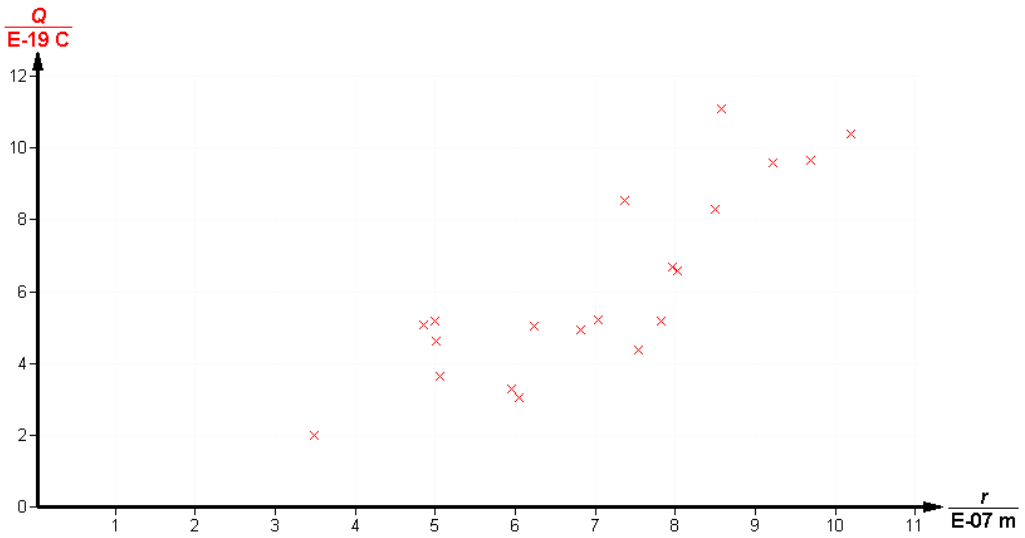


Figure 5: Plot of the Q charges corresponding to the radius r of the oil droplets

10. Draw the lines defining each charge level. The graph should look like Figure 6.

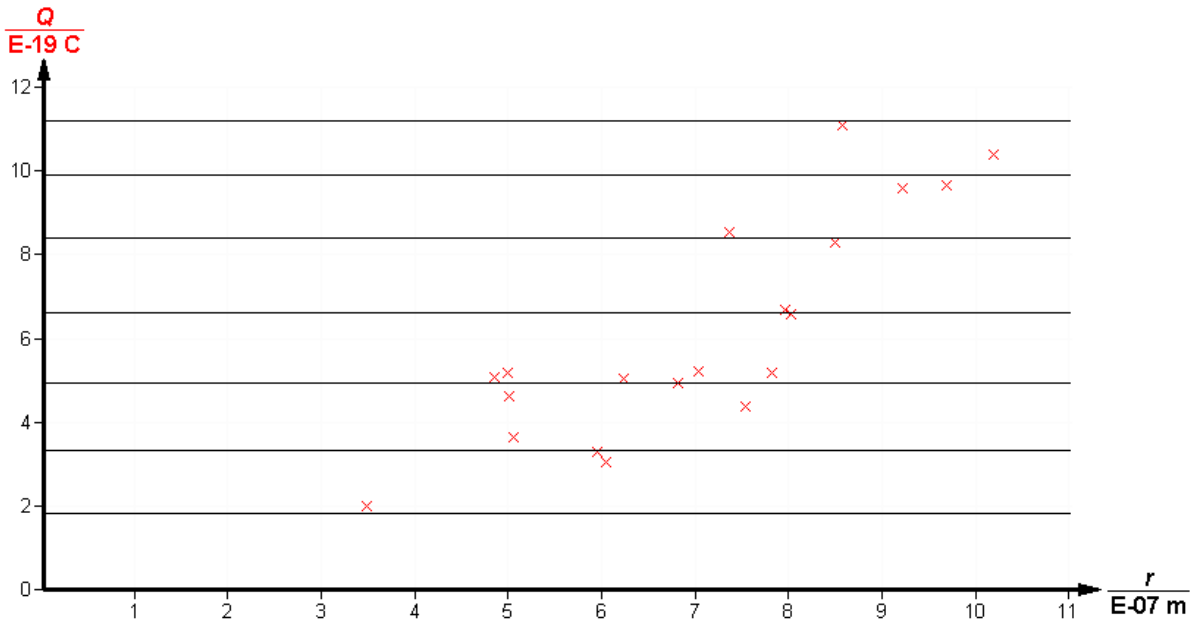


Figure 6

11. Determine the distance between each line. These distances correspond to the elementary charge, *e*. Remember the distance between the x-axis (Q = 0) and the first line. Calculate the distances and tabulate the results in Table 3.

Table 3: Results

Investigated lines	**Difference [C]**
x-axis and first line	
First and second line	
Second and third line	
Third and fourth line	
Fourth and fifth line	
Fifth and sixth line	
Sixth and seventh line	
Average value	

The value of the elementary charge *e* in the literature is $e = 1.602 \times 10^{-19}$ C.

Questions:

1. Explain why charged oil droplets rise or fall.

2. In this experiment several forces act on the charged oil droplet. Draw two graphs: the first graph to show the forces acting on the charged oil droplet as it rises, and the second graph to show the forces acting on the charged oil droplet as it falls.

3. What relation can be observed between the elementary charge and the charge of an oil droplet? Given that N is the number of charges, *e* is the basic charge and Q is the charge of the oil droplet, formulate a formula for the relationship between them.

REFERENCES

1. Catalog of Phywe Elementary Charge and Millikan Experiment
2. www.aip.org